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DETECTING ROTOR DEMAGNETIZATION

Abstract

An electrical power system includes first and second electrical machines, each including permanent magnet rotor and stator including plurality of electrically isolated polyphase winding sets; and a health monitoring system configured to detect rotor demagnetization of first and/or second electrical machine by: supplying electrical power to each polyphase winding set of plurality of polyphase windings sets of both electrical machines; stopping supply of electrical power to a first polyphase winding set of both first and second electrical machines; measuring a first voltage across first polyphase winding set of the first electrical machine; measuring a second voltage across first polyphase winding set of the second electrical machine; determining first and second magnetic flux linkage indicators based on the first and second voltages; and determining whether permanent magnet rotor of first and/or second electrical machine is demagnetized based on a ratio of the first and second magnetic flux linkage indicators.

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Background/Summary

[0001] This disclosure claims the benefit of UK Patent Application No. GB 2403154.4 filed on 5 Mar. 2024, which is hereby incorporated herein in its entirety.

BACKGROUND

Technical Field

[0002] The present disclosure concerns the detection of rotor magnet demagnetization in permanent magnet electrical machines.

Background of the Disclosure

[0003] Permanent magnet synchronous electrical machines (PMSMs) are increasingly used because of their attractive power density. A PMSM generally comprises a rotor that carries permanent magnets, and a stator that comprises a polyphase (e.g., three-phase) winding set. If operating as a motor, the polyphase winding set is energized by an electrical power source, often via an inverter with active power semiconductor switches, to create a rotating magnetic field which interacts with the rotor magnets to create a torque that rotates the rotor. If operating as a generator, the rotor is driven to rotate by a drive shaft (e.g., of a prime mover) and the rotating permanent magnets create a rotating magnetic field that induces a voltage in the stator windings.

[0004] A problem with PMSMs is that the permanent magnets may, over time, become partially or totally demagnetized. This may

happen because of adverse operating conditions, for example high temperatures. Demagnetization of the rotor magnets results in variation in the air-gap magnetic flux density, which may in turn introduce harmonics in the voltage and current waveforms, losses, vibration, acoustic noise and even shaft trajectory displacement. It can also create eddy currents in the permanent magnets, which introduces further losses and can cause significant heating in the magnets, potentially causing further demagnetization.

[0005] Known techniques for detecting rotor demagnetization typically involve the use of complex time- or frequency-domain analysis of measured signals, and the use of threshold values that must be derived from large amounts of experimental or historical operating data. European Patent Publication EP 3876419 B1 derives flux linkage values from current and voltage measurements and compares the flux linkage values to a predefined threshold to detect rotor demagnetization.

SUMMARY

[0006] According to a first aspect there is an electrical power system comprising: [0007] first and second electrical machines, each comprising a permanent magnet rotor and a stator comprising a plurality of electrically isolated polyphase winding sets; and [0008] a control and health monitoring system configured to detect rotor demagnetization of the first and/or second electrical machine by: [0009] supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the first electrical machine; [0010] stopping the supply of electrical power to a first polyphase winding set of the first electrical machine; [0011] measuring a first voltage across the first polyphase winding set of the first electrical machine; [0012] supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the second electrical machine; [0013] stopping the supply of electrical power to a first polyphase winding set of the second electrical machine; [0014] measuring a second voltage across the first polyphase winding set of the second electrical machine; [0015] determining first and second magnetic flux linkage indicators based on the first and second voltages; and [0016] determining whether the permanent magnet rotor of the first and/or second electrical machine is demagnetized based on the whether a ratio of the first and second magnetic flux linkage indicators passes a threshold.

[0017] Optionally, stopping the supply of electrical power to the first polyphase winding set of the first electrical machine comprises opening a circuit breaker connected with the polyphase winding set of the first electrical machine.

[0018] Optionally, the electrical power system further comprises a plurality of DC:AC power converters. For each of the first and second electrical machines, each respective polyphase winding set of the plurality of polyphase winding sets is connected to and configured to receive electrical power from a respective one of the plurality of DC:AC power converters.

[0019] Optionally, stopping the supply of electrical power to the first polyphase winding set of the first electrical machine comprises controlling a switching operation of the respective one of the plurality of DC:AC converters to which the first polyphase winding set of the first electrical machine is connected to block current flow to the first polyphase winding set of the first electrical machine.

[0020] Optionally, supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the first electrical machine comprises supplying less than 25% of a peak rated power of the first electrical machine; and supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the first electrical machine comprises supplying less than 25% of a peak rated power of the second electrical machine.

[0021] Optionally, supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the first electrical machine comprises supplying electrical power to only one phase of a plurality of phases of each polyphase winding set of the first electrical machine, and supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the second electrical machine comprises supplying electrical power to only one phase of a plurality of phases of each polyphase winding set of the second electrical machine.

[0022] Optionally, the electrical power system further comprises an energy storage system, e.g., one or more batteries. Electrical power is supplied to the polyphase winding sets of the first and second electrical machines from the energy storage system.

[0023] Optionally, the energy storage system comprises a plurality of electrically isolated energy storage sub-systems. Each polyphase winding set of the plurality of polyphase winding sets of the first and second electrical machine is configured to receive electrical power from a respective one of the plurality of electrically isolated energy storage sub-systems.

[0024] Optionally, the first and second electrical machines have an equal number of polyphase winding sets.

[0025] Optionally, the number of polyphase winding sets is two or four.

[0026] Optionally, for each of the first and second electrical machines, a circumference of the stator is divided into a plurality of sectors, and each polyphase winding set of the plurality of polyphase winding sets occupies one of the plurality of sectors.

[0027] Optionally, the first and second electrical machines are transverse flux electrical machines.

[0028] Optionally, each polyphase winding set is a three-phase winding set.

[0029] The control and health monitoring system may take any suitable form. Optionally, the control and health monitoring system comprises one or more controllers, for example one or more DC:AC converter controllers. The control and health monitoring system may take any suitable form may be configured to perform a rotor demagnetization detection test as a pre-start check.

[0030] According to a second aspect, there is an aircraft comprising the electrical power system of the first aspect.

[0031] Optionally, the aircraft comprises a first electrical propulsion unit (EPU) comprising a propeller or fan configured to be driven to rotate by the permanent magnet rotor of the first electrical machine; and a second EPU comprising a propeller or fan configured to be driven to rotate by the permanent magnet rotor of the second electrical machine.

[0032] According to a third aspect, there is a method of detecting rotor demagnetization in a permanent magnet electrical machine of an electrical power system, the electrical power system comprising first and second electrical machines, each comprising a permanent magnet rotor and a stator comprising a plurality of electrically isolated polyphase winding sets, [0033] wherein the method comprises:

[0034] supplying electrical power to each polyphase winding set of plurality of polyphase windings sets of the first electrical machine;

[0035] stopping the supply of electrical power to a first polyphase winding set of the first electrical machine; [0036] measuring a first

voltage across the first polyphase winding set of the first electrical machine; [0037] supplying electrical power to each polyphase

winding set of the plurality of polyphase windings sets of the second electrical machine; [0038] stopping the supply of electrical power

to a first polyphase winding set of the second electrical machine; [0039] measuring a second voltage across the first polyphase winding

set of the second electrical machine; [0040] determining first and second magnetic flux linkage indicators based on the first and second

voltages; and [0041] determining whether the permanent magnet rotor of the first and/or second electrical machine is demagnetized

based on the whether a ratio of the first and second magnetic flux linkage indicators passes a threshold.

[0042] Optionally, the method comprises: determining whether the permanent magnet rotor of the first electrical machine is

demagnetized based on the whether the ratio of the first and second magnetic flux linkage indicators is below a first threshold; and

determining whether the permanent magnet rotor of the second electrical machine is demagnetized based on the whether the ratio of the first and second magnetic flux linkage indicators is above a second threshold.

[0043] Optionally, supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the first electrical machine comprises supplying less than 25% of a rated power of the first electrical machine; and supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the first electrical machine comprises supplying less than 25% of a rated power of the first electrical machine.

[0044] Optionally, supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the first electrical machine comprises supplying electrical power to only one phase of a plurality of phases of each polyphase winding set of the first electrical machine; and supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the second electrical machine comprises supplying electrical power to only one phase of a plurality of phases of each polyphase winding set of the second electrical machine.

[0045] Optionally, stopping the supply of electrical power to the first polyphase winding set of the first electrical machine comprises opening a circuit breaker connected with the first polyphase winding set of the first electrical machine is connected.

[0046] Optionally, the electrical power system further comprises a plurality of DC:AC power converters, and, for each of the first and second electrical machines, each respective polyphase winding set of the plurality of polyphase winding sets is connected to and configured to receive electrical power from a respective one of the plurality of DC:AC power converters; and stopping the supply of electrical power to the first polyphase winding set of the first electrical machine comprises controlling a switching operation of the DC:AC converter to which the first polyphase winding set of the first electrical machine is connected to block current flow to the first polyphase winding set of the first electrical machine.

[0047] The skilled person will appreciate that, except where mutually exclusive, a feature described in relation to any one of the above aspects may be applied mutatis mutandis to any other aspect. Furthermore, except where mutually exclusive, any feature described herein may be applied to any aspect and/or combined with any other feature described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0048] Embodiments will now be described by way of example only with reference to the accompanying drawings, which are purely schematic and not to scale, and in which:

[0049] FIG. 1 is a perspective view of a VTOL aircraft;

[0050] FIG. 2 is a schematic illustration of a power and propulsion system of an aircraft comprising a gas turbine engine;

[0051] FIG. 3A illustrates a permanent magnet radial flux electrical machine comprising two electrically isolated polyphase winding sets;

[0052] FIG. 3B illustrates a permanent magnet transverse flux electrical machine comprising two electrically isolated polyphase winding sets;

[0053] FIG. 4 is a schematic illustration of a portion of an electrical power system, configured to detect rotor demagnetization in its permanent magnet electrical machines;

[0054] FIG. 5 is a schematic illustration of a portion of an electrical power system, in which the permanent magnet electrical machines have four electrically isolated polyphase winding sets; and

[0055] FIG. 6 is a flow chart illustrating a method of operating an electrical power system to detect rotor demagnetization.

DETAILED DESCRIPTION

[0056] FIG. 1 illustrates a Vertical Take-Off and Landing (VTOL) aircraft 1. The VTOL aircraft has a fuselage 10, a front flight surface 11, a rear flight surface 12 and a plurality of electrical propulsion units (EPUs). In this example, the VTOL aircraft 1 has six EPUs, including four front EPUs 10f coupled to the front flight surface 11 and two rear EPUs 10r coupled to the rear flight surface 12. Each EPU 10f, 10r includes a propeller 14, which could instead be a fan (e.g., a ducted fan). Some or all of the EPUs 10f, 10r are tiltable between a vertical take-off configuration (illustrated) and a horizontal flight configuration. In this example, the front and rear flight surfaces 11, 12 are tiltable and the EPUs 10f, 10r tilt therewith. In other examples, the EPUs 10f, 10r may instead be tiltable relative to the flight surfaces 11, 12.

[0057] The propeller 14 of each EPU 10f, 10r can be driven to rotate by an electrical machine (not visible) incorporated within the EPU. The electrical machines receive power from an on-board power source (also not visible), which in one example is an Energy Storage System (ESS) comprising one or more batteries. In other examples, the VTOL aircraft 1 has a hybrid power system that includes, for example, batteries and fuel cells, or batteries and a turboelectric generator (e.g., a gas turbine engine and electrical machine coupled to a rotating shaft of the gas turbine engine).

[0058] FIG. 2 is a schematic illustration of a power and propulsion system 2 of an aircraft that includes a gas turbine engine 20. In this example the gas turbine engine 20 has two spools, a high-pressure (HP) spool and a low-pressure (LP) spool. The HP spool is mechanically coupled to a first electrical machine 21 and the LP spool is coupled to a second electrical machine 22. The electrical machines 21, 22 interface with a first DC electrical network 25 via DC:AC power converters 23, 24, which in this example are bi-directional DC:AC power converters whereby the electrical machines are operable as generators (for supplying power to the first DC electrical network 25) or motors (for accelerating the LP and/or HP spool, e.g., for engine starting or improving engine operability).

[0059] The first DC electrical network 25, which in this example powers electrical loads associated with the gas turbine engine 20 (e.g., electric engine accessories such as motors of fuel or oil pumps), can also exchange power with an ESS 26, optionally via a DC:DC power converter 27. The power and propulsion system further includes a second DC electrical network 28 that powers platform loads (i.e., those associated with the wider aircraft). The voltage of the second electrical network 28 is different than (e.g., less than) the voltage of the first electrical network 25, and a second DC:DC power converter 29 provides an interface between the two DC electrical networks 25, 28.

[0060] A control system 30, which may comprise single controller or multiple distributed controllers and various sensors, may control the operating modes of the electrical machines 21, 22 and the switching operation of the power semiconductor switches of the power converters 23, 24, 27, 29. In accordance with the present disclosure, the control system 30 may also perform health monitoring

operations to detect rotor demagnetization in electrical machines **21**, **22**.

[0061] Electrical power systems used in aerospace applications such as those described above may have relatively high fault tolerance requirements. To this end, the electrical machines may be designed to have redundancy. One example of this is the provision of permanent magnet electrical machines that have stators with multiple electrically independent stator winding sets that interact with a common rotor. Non-limiting examples of this are illustrated in FIGS. 3A-3B.

[0062] FIG. 3A illustrates a permanent magnet radial flux electrical machine **40**. The electrical machine **40** has a rotor **41** and a stator **42** that radially surrounds the rotor, though the rotor **41** could instead radially surround the stator **42**. A surface of the rotor **41** facing the stator **42** carries permanent magnets (e.g., rare earth magnets) for interaction with the stator **42**. The stator **42** has a generally annular back iron **421** and a plurality of teeth **422** that extend radially from the back iron **421** and are evenly distributed about the circumference of the stator **42**. The stator **42** further includes two electrically independent three-phase winding sets, the coils of which are wound around the teeth **422**. In FIG. 3A the two winding sets are designated “1” and “2”, and the three phases are designated “U”, “V” and “W”. Thus, the first three-phase winding set has phases U1, V1, W1 and these are associated with the coils **423**, **425**, **427**. The second winding set has phases U2, V2, W2 and these are associated with the coils **424**, **426**, **428**.

[0063] The coils **423**, **425**, **427** that form the first winding set are electrically connected, for example in a star or delta configuration. Likewise, the coils **424**, **426**, **428** that form the second winding set are electrically connected, for example in a star or delta configuration. However, to ensure the first and second winding sets are electrically isolated, the first set of coils **423**, **425**, **427** are not electrically connected with the second set of coils **424**, **426**, **428**. Thus, if there is an electrical fault (e.g., a phase fault) in one of the winding sets, the fault should not electrically propagate to the other winding set. A fault in one winding set that, e.g., requires disconnection of the winding set from the wider power system does not prevent the motor **40** from operating, and with suitable overrating of the motor **40**, it may continue operation at full or close to full rated power.

[0064] In the present example, the circumferentially adjacent coils (e.g., coils **423**, **424**) belong to different winding sets. In other examples, including those shown in FIG. 3B and FIG. 5, the circumference of the stator is divided into arcuate sectors, the angular size of each sector being $360^\circ/N$ where N is the number of winding sets, and each winding set occupies one of the N sectors. In the present example, a concentrated winding scheme is applied, and a coil is wound around every second stator tooth **422** so that an unwound tooth separates circumferentially adjacent coils. This is not essential, but increases the thermal, mechanical and magnetic isolation between the two winding sets and thus further reduces the risk of fault propagation.

[0065] FIG. 3B illustrates a permanent magnet transverse flux electrical machine **50**. The electrical machine **50** has a permanent magnet rotor **51**, which in this example comprises first and second rotor portions **51a**, **51b**. The electrical machine **50** further includes a stator **52** provided between the rotor portions **51a**, **51b**. Each of the rotor portions **51a**, **51b** carries an array of permanent magnets (e.g., a Halbach array) on a side of the rotor portion facing the stator **52** for interaction therewith. The stator **52** comprises two three-phase winding sets. As in FIG. 3A, the two winding sets are designated “1” and “2”, and the three phases are designated “U”, “V” and “W”.

[0066] In this example, the rotor portions **51a**, **51b** are radially spaced and the stator **52** is provided radially between the rotor portions **51a**, **51b**. In other transverse flux electrical machines, the rotor portions **51a**, **51b** are axially spaced and the stator **52** is provided axially between the rotor portions **51a**, **51b**. The use of two rotor portions **51a**, **51b** helps close the magnetic circuits without the requirement for rotor iron (i.e., an ironless rotor can be used), but this is not essential; some transverse flux electrical machines have only a single rotor portion.

[0067] Each phase (U, V, W) of each winding set (1, 2) of the stator **52** is associated with a stator segment **521-526**, and each stator segment **521-526** comprises a set of flux guiding iron pieces (e.g., comprising a soft magnetic composite) and a coil through which coil flows in a generally circumferential direction. It is noted that FIG. 3B shows the flux guiding iron pieces but omits the coils for clarity. Further examples may be found in US Patent Publication U.S. Pat. No. 11,611,247 B2, the entire contents of which are incorporated herein by reference.

[0068] The coils **521**, **522**, **523** that form the first winding set are electrically connected, for example in a star or delta configuration. Likewise, the coils **524**, **525**, **526** that form the second winding set are electrically connected, for example in a star or delta configuration. However, to ensure the first and second winding sets are electrically isolated, the first set of coils **521-523** are not electrically connected with the second set of coils **524-526**. Thus, if there is an electrical fault (e.g., a phase fault) in one of the winding sets, the fault should not electrically propagate to the other winding set. A fault in one winding set that, e.g., requires disconnection of the winding set does not prevent the motor **50** from operating, and with suitable overrating of the motor **50**, it may continue operation at full or close to full power.

[0069] The circumference of the stator is divided into sectors **52-1**, **52-2**, the angular size of each sector being $360^\circ/N$ where N is the number of winding sets (here N=2), and each winding set occupies one of the sectors. In other examples, analogous to that shown in FIG. 3A, circumferentially adjacent stator segments (e.g., stator segments **521**, **522**) are associated with different winding sets.

[0070] In each of the examples described above, the stators **42**, **52** have two electrically independent winding sets each having three-phases. A different number of phases (e.g., four) could be used instead and the term “polyphase” is used herein to indicate this. Further, a stator could have more than two (e.g., three or four) electrically isolated winding sets. The number of winding sets and phases may be selected according to the desired level of fault tolerance, as well as complexity and space constraints, for example.

[0071] FIG. 4 illustrates an electrical power system **100** that includes two permanent magnet electrical machines **110**, **120**. The electrical power system **100** could, for example, be a portion of that of the VTOL aircraft **1** of FIG. 1, with the two electrical machines **110**, **120** being motors of two propulsors **10f**, **10r**. The electrical power system **100** could also be that of the FIG. 2, with the two electrical machines **110**, **120** being the electrical machines **21**, **22** that are coupled to the spools of the gas turbine engine **20**. The electrical machines **110**, **120** could be permanent magnet radial flux machines, like that described with reference to FIG. 3A.

Alternatively, the electrical machines **110**, **120** could be permanent magnet transverse flux machines like that described with reference to FIG. 3B, or another type of permanent magnet machine (e.g., axial flux machines).

[0072] The first permanent magnet electrical machine **110** has a first winding set **110-1** having three phases **110-1U**, **110-1V**, **110-1W**, and a second winding set **110-2** having three phases **110-2U**, **110-2V**, **110-2W**. The second permanent magnet electrical machine **120** has a first winding set **120-1** having three phases **120-1U**, **120-1V**, **120-1W**, and a second winding set **120-2** having three phases **120-2U**, **120-2V**, **120-2W**. As noted above, a different number of phases may be used.

[0073] In the present example, each winding set **110-1**, **110-2**, **120-1**, **120-2** is connected to one of a plurality of DC:AC power converters **130-1**, **130-2**, **140-1**, **140-2**. The DC:AC power converters **130-1**, **130-2**, **140-1**, **140-2** may be unidirectional (e.g., the

power converters are inverters) or bidirectional. Any suitable converter topology may be used, for example a two-level three-phase full-bridge converter topology, or each DC:AC power converter may comprise three one-phase H-bridge circuits and each phase may be connected to one of the three H-bridges. In still other examples, the DC:AC converters may not be present at all, for example, if the electrical machines **110**, **120** are to be supplied without power from an AC power source.

[0074] The DC:AC converters **130-1**, **130-2**, **140-1**, **140-2** are connected, at their DC sides, to a DC power source **160**, which in this example is an ESS, for example an on-board ESS of an aircraft. In other examples the power source **160** may be a ground-based power source (e.g., airport ground equipment) or an on-board Auxiliary Power Unit (APU).

[0075] In the present example, the ESS **160** comprises, for each polyphase winding set **110-1**, **110-2**, **120-1**, **120-2** and associated DC:AC power converter **130-1**, **130-2**, **140-1**, **140-2**, an electrically isolated energy storage sub-system **161**, **162**, **163**, **164**. The combination of an energy storage sub-system (e.g., **161**), the associated DC:AC converter (e.g., **130-1**) and the associated polyphase stator winding (e.g., **110-1**) may be referred to as a “power lane”. Each power lane is electrically isolated from every other power lane, which reduces the likelihood of faults propagating between power lanes and therefore the impact of any single fault.

[0076] The electrical power system **100** further includes a control and health monitoring system **150**, which is communicatively coupled with voltage sensors **151**, **152**. The control and health monitoring system **150** is configured to detect the onset of rotor demagnetization in the permanent magnet rotors of the electrical machines **110**, **120**. To do so, the control and health monitoring system **150** initially supplies power to all the winding sets **110-1**, **110-2**, **120-1**, **120-2** of the two electrical machines **110**, **120** to spin up their permanent magnet rotors. The control and health monitoring system **150** then imposes an open circuit condition on one winding set of each electrical machine **110**, **120**, after which the open circuit winding sets can behave like search coils. Voltages across the open circuit winding sets are measured, and magnetic flux linkage indicators calculated from the measurements. If the two electrical machines **110**, **120** are of the same construction and are subject to the measurement conditions during the test, a ratio of the flux linkage indicators will indicate whether there is demagnetization in one of the rotors. For example, a value close to one will indicate there is little or no rotor demagnetization, whereas a value deviating from close to one will indicate one of the rotors has some demagnetization. The rotor demagnetization test is described in more detail below.

[0077] First, the control and health monitoring system **150** controls the power system **100** (e.g., controls the switching states of the DC:AC power converters **130-1**, **130-2**, **140-1**, **140-2**) to supply electrical power to each of the polyphase winding sets **110-1**, **110-2**, **120-1**, **120-2** of the two electrical machines **110**, **120** so that their rotors rotate.

[0078] The same voltage and current may be supplied to each machine **110**, **120** so that the rotors rotate at the same speed and produce the same torque. For the purposes of the rotor demagnetization test, the electrical machines **110**, **120** may not be operated at their full powers, speeds or torques, and thus a relatively low power (e.g., less than 25% of the peak rated powers of the machines **110**, **120**) may be supplied to the polyphase winding sets **110-1**, **110-2**, **120-1**, **120-2**. In one group of examples, power is not supplied to every phase of the winding sets. For example, power may be supplied to just one phase of each winding set (e.g., **110-1U**, **110-2V**, **120-1U**, **120-2V**) or to two phases of each winding set.

[0079] Next, the control and health monitoring system **150** stops the supply of electrical power to one of the polyphase winding sets of each of the first and second electrical machines **110**, **120**, e.g., polyphase winding sets **110-2**, **120-2**. The supply of current to the other polyphase winding sets **110-1**, **120-1** continues.

[0080] In the illustrated example, the supply of electrical power to the winding sets **110-2**, **120-2** is stopped by opening circuit breakers **171**, **172** that are connected between the winding sets **110-2**, **120-2** and their associated DC:AC converters **130-2**, **140-2**, disconnecting the winding sets **110-2**, **120-2**. In another example, the control and health monitoring system **150** controls the switching states of the DC:AC power converters **130-2**, **140-2** to block the flow of current to the winding sets **110-2**, **120-2**. Where circuit breakers **171**, **172** are used, they may be mechanical circuit breakers (e.g., AC contactors), solid state circuit breakers (SSCBs) or hybrid breakers.

[0081] With the supply of current to the winding sets **110-2**, **120-2** stopped but the permanent magnet rotors still rotating, voltages will be induced in the unpowered winding sets **110-2**, **120-2**. The voltage sensors **151**, **152** that are connected across the winding sets **110-2**, **120-2** measure the phase voltages across the winding sets and communicate the measurements to the control and health monitoring system **150** for processing.

[0082] The voltages $V_{sub.12}$, $V_{sub.22}$ are related to the magnetic flux linkages in the winding sets by the equations:

[00001]

$$\begin{bmatrix} v_{u12} \\ v_{v12} \\ v_{w12} \end{bmatrix} = \begin{bmatrix} R_{s12} & 0 & 0 \\ 0 & R_{s12} & 0 \\ 0 & 0 & R_{s12} \end{bmatrix} \begin{bmatrix} i_{u12} \\ i_{v12} \\ i_{w12} \end{bmatrix} + \begin{bmatrix} \frac{d}{dt} \frac{\psi_{u12}}{dt} \\ \frac{d}{dt} \frac{\psi_{v12}}{dt} \\ \frac{d}{dt} \frac{\psi_{w12}}{dt} \end{bmatrix} = \begin{bmatrix} \frac{d}{dt} \frac{\psi_{u12}}{dt} \\ \frac{d}{dt} \frac{\psi_{v12}}{dt} \\ \frac{d}{dt} \frac{\psi_{w12}}{dt} \end{bmatrix} \begin{bmatrix} v_{u22} \\ v_{v22} \\ v_{w22} \end{bmatrix} = \begin{bmatrix} R_{s22} & 0 & 0 \\ 0 & R_{s22} & 0 \\ 0 & 0 & R_{s22} \end{bmatrix} \begin{bmatrix} i_{u22} \\ i_{v22} \\ i_{w22} \end{bmatrix} + \begin{bmatrix} \frac{d}{dt} \frac{\psi_{u22}}{dt} \\ \frac{d}{dt} \frac{\psi_{v22}}{dt} \\ \frac{d}{dt} \frac{\psi_{w22}}{dt} \end{bmatrix} = \begin{bmatrix} \frac{d}{dt} \frac{\psi_{u22}}{dt} \\ \frac{d}{dt} \frac{\psi_{v22}}{dt} \\ \frac{d}{dt} \frac{\psi_{w22}}{dt} \end{bmatrix}$$

Here, $v_{sub.u12}$ is the phase voltage associated with phase u for the winding set **110-2**, $v_{sub.w22}$ is the phase voltage associated with phase w for the winding set **120-2**, and so forth—these values are measured by voltage sensors **151**, **152**. $\psi_{sub.u12}$ is the magnetic flux linkage of the phase u of the winding set **110-2**, $\psi_{sub.w22}$ is the magnetic flux linkage of the phase w of the winding set **120-2**, and so forth.

[00002] $\frac{d}{dt} \frac{\psi_{u12}}{dt}$

and

[00003] $\frac{d}{dt} \frac{\psi_{w22}}{dt}$

are their time derivatives. $R_{sub.s12}$ and $R_{sub.s22}$ are the equivalent resistances of the stator winding sets **110-2**, **120-2**, while the currents, e.g., $i_{sub.u12}$, are the individual phase currents. The winding sets **110-2**, **120-2** are in an open circuit state when the voltages are measured, so the phase currents are zero, and so the intermediate term is also zero. Consequently, integrating the measured voltages with respect to time will give flux linkage values.

[0083] Thus, after receiving the measurements of the phase voltages, the control and health monitoring system **150** integrates the phase voltage values to give the magnetic flux linkage values. The control and health monitoring system **150** then determines a flux value indicator, $F_{sub.1}$, $F_{sub.2}$ from the flux linkage values. For example, the control and health monitoring system **150** may take the absolute values of the flux linkage vectors to give the indicators:

$$[00004] F_1 = \sqrt{\psi_{u12}^2 + \psi_{v12}^2 + \psi_{w12}^2} \quad F_2 = \sqrt{\psi_{u22}^2 + \psi_{v22}^2 + \psi_{w22}^2}$$

Another flux linkage indicator value could be calculated, provided the same calculation is used for both $F_{sub.1}$ and $F_{sub.2}$.

[0084] Next, the control and health monitoring system **150** determines a ratio, $F_{\text{sub.1}}/F_{\text{sub.2}}$. If the ratio is approximately equal to one, this indicates that neither of the electrical machines has significant rotor demagnetization. If the ratio is less than one (e.g., less than a first predefined threshold), this indicates the rotor of the first electrical machine has at least some rotor demagnetization. If the ratio is greater than one (e.g., greater than a second predefined threshold), this indicates the rotor of the second electrical machine has at least some rotor demagnetization. In one specific example, the first threshold is 0.9 and the second threshold is 1.1. In another specific example, first threshold is 0.8 and the second threshold is 1.2.

[0085] Notably, since the ratio is a dimensionless value and the electrical machines **110**, **120** are of the same construction and subject to the same test conditions, thresholds can be defined without recourse to extensive testing or historical operating data. Further, the level of complexity associated with performing the test is relatively low.

[0086] FIG. 5 illustrates an example electrical power system **100** in which each of the electrical machines **110**, **120** has four electrically isolated polyphase winding sets. Specifically, the first electrical machine **110** has a first three-phase winding set **110-1**, a second three-phase winding set **110-2**, a third three-phase winding set **110-3** and a fourth three-phase winding set **110-4**. The second electrical machine **120** has a first three-phase winding set **120-1**, a second three-phase winding set **120-2**, a third three-phase winding set **120-3** and a fourth three-phase winding set **120-4**. In this example, the circumferences of the stators are divided into $N=4$ sectors, each subtending an angle of 90° , and each polyphase winding set occupies one of the sectors.

[0087] As in FIG. 4, each polyphase winding set **110-1**, **110-2**, **110-3**, **110-4**, **120-1**, **120-2**, **120-3**, **120-4** is connected to one of a plurality of DC:AC power converters **130-1**, **130-2**, **130-3**, **130-4**, **140-1**, **140-2**, **140-3**, **140-4**. As before, the DC:AC power converters may be of any suitable topology and may be omitted entirely if an AC power source is used. It is noted that, for clarity, no power sources (e.g., ESS **160**) are shown in FIG. 5.

[0088] The electrical power system **100** further includes a control and health monitoring system **150**, voltage sensors **151**, **152** and circuit breakers **171**, **172**. The control and health monitoring system **150** is configured to detect the onset of rotor demagnetization in the permanent magnet electrical machines **110**, **120**. The details of the rotor demagnetization test are substantially the same as described above. Initially, power is supplied to each of the winding sets **110-1**, **110-2**, **110-3**, **110-4**, **120-1**, **120-2**, **120-3**, **120-4** of the electrical machines. One of the winding sets of each machine, e.g., winding sets **130-1**, **140-1**, is then placed in an open circuit condition, for example by opening the circuit breakers **171**, **172**. The voltage sensors **151**, **152** then measure the phase voltages across each of the open circuit winding sets **130-1**, **140-1**. The control and health monitoring system **150** receives the voltage measurements and determines magnetic flux indicators **F1**, **F2** from the voltage measurements. Whether there is rotor demagnetization is determined based on whether a ratio of the flux indicators passes a threshold.

[0089] FIG. 6 is a flow chart illustrating a method **200** of operating an electrical power system **100** to detect rotor demagnetization. The electrical power system **100**, which may be of the types shown in FIGS. 4-5, comprises at least two electrical machines, each comprising at least two electrically isolated polyphase winding sets. The method **200** may be performed under the control of a control and health monitoring system **150**, which may be pre-programmed to perform a rotor demagnetization test, e.g., during a prestart system check. In an example, the control and health monitoring system **150** may be or comprise a power converter controller.

[0090] At **201**, the control and health monitoring system **150** initiates the supply of electrical power to all of the winding sets of the first electrical machine **110** (e.g., **110-1**, **110-2** in FIG. 4). In an example, the control and health monitoring system **150** controls the DC:AC converters (e.g., **130-1**, **130-2**) to initiate the supply of current to the windings. In some examples, power is supplied to all of the phases of the winding set. In another example, current is supplied to a subset (e.g., one) of the phases of each winding set. A low power (e.g., less than 25% of the rated power) sufficient to rotate the rotor may be supplied to electrical machine **110**.

[0091] At **202**, the control and health monitoring system **150** stops the supply of electrical power to one of the winding sets of the first electrical machine (e.g., winding set **110-2** in FIG. 4). In an example, the control and health monitoring system **150** controls the DC:AC converter **130-2** to stop the flow of current therethrough and opens the contactor **171**, placing the winding set **110-2** in an open circuit condition.

[0092] At **203**, the voltage sensor **151** measures the phase voltages across the open circuit winding set **110-2** and communicates the voltages to the control and health monitoring system **150** for processing.

[0093] At **204**, the control and health monitoring system **150** determined a first magnetic flux linkage indicator, **F1**, for the first electrical machine **110**. In an example, the control and health monitoring system **150** integrates the phase voltage to give magnetic flux linkage values and then calculates **F1** as a predefined function (e.g., the absolute value) of the magnetic flux linkage values.

[0094] At **205**, the control and health monitoring system **150** stops the supply of electrical power to the first electrical machine **110**.

[0095] At **206**, the control and health monitoring system **150** initiates the supply of electrical power to all of the winding sets of the second electrical machine **120** (e.g., **120-1**, **120-2** in FIG. 4). In an example, the control and health monitoring system **150** controls the DC:AC converters (e.g., **140-1**, **140-2**) to initiate the supply of current to the windings. In some examples, power is supplied to all of the phases of the winding set. In another example, current is supplied to a subset (e.g., one) of the phases of each winding set. A low power (e.g., less than 25% of the rated power) sufficient to rotate the rotor may be supplied to electrical machine **120**.

[0096] At **207**, the control and health monitoring system **150** stops the supply of electrical power to one of the winding sets of the second electrical machine (e.g., winding set **120-2** in FIG. 4). In an example, the control and health monitoring system **150** controls the DC:AC converter **140-2** to stop the flow of current therethrough and opens the contactor **172**, placing the winding set **120-2** in an open circuit condition.

[0097] At **208**, the voltage sensor **152** measures the phase voltages across the open circuit winding set **120-2** and communicates the voltages to the control and health monitoring system **150** for processing.

[0098] At **209**, the control and health monitoring system **150** determined a second magnetic flux linkage indicator, **F2**, for the second electrical machine **120**. In an example, the control and health monitoring system **150** integrates the phase voltage to give magnetic flux linkage values and then calculates **F2** as a predefined function (e.g., the absolute value) of the magnetic flux linkage values.

[0099] At **210**, the control and health monitoring system **150** stops the supply of electrical power to the first electrical machine **120**.

[0100] At **211**, the control and health monitoring system **150** calculates a ratio, $F1/F2$, of the first and second flux linkage indicators.

[0101] At **212**, the control and health monitoring system **150** asks the question of whether the ratio is less than the first predetermined threshold. In an example, the first threshold has a value of between 0.8 and 0.9.

[0102] If the ratio is less than the first predetermined threshold, at **213** the control and health monitoring system **150** determines that the first electrical machine **110** is subject to rotor demagnetization. The control and health monitoring system **150** may output an

indicator this effect, for example to an aircraft cockpit display.

[0103] At **214**, the control and health monitoring system **150** asks the question of whether the ratio is greater than a second predetermined threshold. In an example, the second threshold has a value of between 1.1 and 1.2.

[0104] If the ratio is less than the first predetermined threshold, at **215** the control and health monitoring system **150** determines that the second electrical machine **120** is subject to rotor demagnetization. The control and health monitoring system **150** may output an indicator this effect, for example to an aircraft cockpit display.

[0105] Embodiments of the present disclosure may be associated with one or more of the following advantages: [0106] The threshold(s) utilized by the control and health monitoring system **100** may be selected without requiring extensive test and/or historical data. [0107] Rotor demagnetization may be detected without the addition of many, if any, sensors to the electrical power system **100**.

[0108] It is not necessary for the sensors to continuously measure the voltage (and current) with a high sampling rate. [0109] The method may be applied to any type of PMSM, irrespective of the type and winding arrangement selected for the polyphase winding sets. [0110] The processing complexity is relatively low and may not require a high-end processor to execute complex algorithms.

[0111] Reduced system downtime and increased reliability.

[0112] It will be understood that the invention is not limited to the embodiments above-described and various modifications and improvements can be made without departing from the concepts herein. The invention has been described with reference to aerospace applications but could be used in other transport and non-transport applications. Except where mutually exclusive, any of the features may be employed separately or in combination with any other features and the disclosure extends to and includes all combinations and sub-combinations of one or more features described herein.

[0113] Various examples have been described, each of which feature various combinations of features. It will be appreciated by those skilled in the art that, except where clearly mutually exclusive, any of the features may be employed separately or in combination with any other features and the invention extends to and includes all combinations and sub-combinations of one or more features described herein.

Claims

1. An electrical power system comprising: first and second electrical machines, each comprising a permanent magnet rotor and a stator comprising a plurality of electrically isolated polyphase winding sets; and a control and health monitoring system configured to detect rotor demagnetization of the first and/or second electrical machine by: supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the first electrical machine; stopping the supply of electrical power to a first polyphase winding set of the first electrical machine; measuring a first voltage across the first polyphase winding set of the first electrical machine; supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the second electrical machine; stopping the supply of electrical power to a first polyphase winding set of the second electrical machine; measuring a second voltage across the first polyphase winding set of the second electrical machine; determining first and second magnetic flux linkage indicators based on the first and second voltages; and determining whether the permanent magnet rotor of the first and/or second electrical machine is demagnetized based on the whether a ratio of the first and second magnetic flux linkage indicators passes a threshold.
2. The electrical power system of claim 1, wherein stopping the supply of electrical power to the first polyphase winding set of the first electrical machine comprises opening a circuit breaker connected with the polyphase winding set of the first electrical machine.
3. The electrical power system of claim 1, further comprising a plurality of DC:AC power converters, wherein, for each of the first and second electrical machines, each respective polyphase winding set of the plurality of polyphase winding sets is connected to and configured to receive electrical power from a respective one of the plurality of DC:AC power converters.
4. The electrical power system of claim 3, wherein stopping the supply of electrical power to the first polyphase winding set of the first electrical machine comprises controlling a switching operation of the respective one of the plurality of DC:AC converters to which the first polyphase winding set of the first electrical machine is connected to block current flow to the first polyphase winding set of the first electrical machine.
5. The electrical power system of claim 1, wherein: supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the first electrical machine comprises supplying less than 25% of a peak rated power of the first electrical machine; and supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the second electrical machine comprises supplying less than 25% of a peak rated power of the second electrical machine.
6. The electrical power system of claim 1, wherein supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the first electrical machine comprises supplying electrical power to only one phase of a plurality of phases of each polyphase winding set of the first electrical machine, and supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the second electrical machine comprises supplying electrical power to only one phase of a plurality of phases of each polyphase winding set of the second electrical machine.
7. The electrical power system of claim 1, further comprising an energy storage system, wherein electrical power is supplied to the polyphase winding sets of the first and second electrical machines from the energy storage system.
8. The electrical power system of claim 7, wherein the energy storage system comprises a plurality of electrically isolated energy storage sub-systems, and wherein each polyphase winding set of the plurality of polyphase winding sets of the first and second electrical machine is configured to receive electrical power from a respective one of the plurality of electrically isolated energy storage sub-systems.
9. The electrical power system of claim 1, wherein the first and second electrical machines have an equal number of polyphase winding sets.
10. The electrical power system of claim 9, wherein the number of polyphase winding sets is two or four.
11. The electrical power system of claim 1, wherein, for each of the first and second electrical machines, a circumference the stator is divided into a plurality of sectors and each polyphase winding set of the plurality of polyphase winding sets occupies one of the plurality of sectors.
12. The electrical power system of claim 1, wherein the first and second electrical machines are transverse flux electrical machines.
13. An aircraft comprising the electrical power system of claim 1.

14. The aircraft of claim 13, comprising: a first electrical propulsion unit comprising a propeller or fan configured to be driven to rotate by the permanent magnet rotor of the first electrical machine; and a second electrical propulsion unit comprising a propeller or fan configured to be driven to rotate by the permanent magnet rotor of the second electrical machine.

15. A method of detecting rotor demagnetization in a permanent magnet electrical machine of an electrical power system, the electrical power system comprising first and second electrical machines, each comprising a permanent magnet rotor and a stator comprising a plurality of electrically isolated polyphase winding sets, wherein the method comprises: supplying electrical power to each polyphase winding set of plurality of polyphase windings sets of the first electrical machine; stopping the supply of electrical power to a first polyphase winding set of the first electrical machine; measuring a first voltage across the first polyphase winding set of the first electrical machine; supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the second electrical machine; stopping the supply of electrical power to a first polyphase winding set of the second electrical machine; measuring a second voltage across the first polyphase winding set of the second electrical machine; determining first and second magnetic flux linkage indicators based on the first and second voltages; and determining whether the permanent magnet rotor of the first and/or second electrical machine is demagnetized based on the whether a ratio of the first and second magnetic flux linkage indicators passes a threshold.

16. The method of claim 15, comprising: determining whether the permanent magnet rotor of the first electrical machine is demagnetized based on the whether the ratio of the first and second magnetic flux linkage indicators is below a first threshold; and determining whether the permanent magnet rotor of the second electrical machine is demagnetized based on the whether the ratio of the first and second magnetic flux linkage indicators is above a second threshold.

17. The method of claim 15, wherein: supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the first electrical machine comprises supplying less than 25% of a peak rated power of the first electrical machine; and supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the second electrical machine comprises supplying less than 25% of a peak rated power of the second electrical machine.

18. The method of claim 15, wherein: supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the first electrical machine comprises supplying electrical power to only one phase of a plurality of phases of each polyphase winding set of the first electrical machine; and supplying electrical power to each polyphase winding set of the plurality of polyphase windings sets of the second electrical machine comprises supplying electrical power to only one phase of a plurality of phases of each polyphase winding set of the second electrical machine.

19. The method of claim 15, wherein stopping the supply of electrical power to the first polyphase winding set of the first electrical machine comprises opening a circuit connected with the first polyphase winding set of the first electrical machine is connected.

20. The method of claim 15, wherein: the electrical power system further comprises a plurality of DC:AC power converters, and, for each of the first and second electrical machines, each respective polyphase winding set of the plurality of polyphase winding sets is connected to and configured to receive electrical power from a respective one of the plurality of DC:AC power converters; and stopping the supply of electrical power to the first polyphase winding set of the first electrical machine comprises controlling a switching operation of the DC:AC converter to which the first polyphase winding set of the first electrical machine is connected to block current flow to the first polyphase winding set of the first electrical machine.
